

Week1_Baseline Qualitative Analysis

Objective: To compare the reasoning capabilities of selected language models using qualitative analysis and identify common reasoning transition markers.

Q1:	Arithmetic Reasoning				
QUESTION	A Shopkeeper buys a abook for Rs.250 and sells it for Rs.325.What is the profit?				
Model	Observation	Final Answer	Correct	Reasoning Markers	Reasoning Quality
Meta Llama 3 8B	Detailed and accurate reasoning	Rs.75	✓	First,Therefore	Excellent
Google Gemma 2 9B	Clear and logical explanation	Rs.75	✓	Because,Hence	Excellent
Mistral 7B	Correct but concise	Rs.75	✓	Therefore	Very Good
Microsoft Phi-3 Mini	Short explanation	Rs.75	✓	So	Very Good
Qwen2.5 7B	Excellent mathematical reasoning	Rs.75	✓	First,Therefore,Because	Excellent
Google Gemma 3 4B	Correct with brief reasoning	Rs.75	✓	Next,Finally	Very Good
Q2:	Logical Reasoning				
QUESTION	All roses are flowers.Some flowers fade quickly.Can we conclude that some roses fade quickly?				
Model	Observation	Final Answer	Correct	Reasoning Markers	Reasoning Quality
Meta Llama 3 8B	Correct logical deduction	No	✓	Because, Therefore	Excellent
Google Gemma 2 9B	Well-structured explanation	No	✓	Hence	Excellent
Mistral 7B	Concise but correct	No	✓	Therefore	Very Good
Microsoft Phi-3 Mini	Correct logical assumption	No	✓	Because	Fair
Qwen2.5 7B	Excellent reasoning	No	✓	First,Therefore	Excellent
Google Gemma 3 4B	Good explanation	No	✓	Hence	Very Good
Q3:	Multi-step Math Reasoning				
QUESTION	A train travels at 60 km/h for 2 hours and then at 80 km/h for 3 hours.What is the total distance travelled?				
Model	Observation	Final Answer	Correct	Reasoning Markers	Reasoning Quality
Meta Llama 3 8B	Detailed calculation	360km	✓	First,Next,Therefore	Excellent
Google Gemma 2 9B	Accurate and structured	360km	✓	Because,Finally	Excellent
Mistral 7B	Correct but brief	360km	✓	Therefore	Very Good
Microsoft Phi-3 Mini	Correct calculation	360km	✓	So	Fair
Qwen2.5 7B	Excellent reasoning	360km	✓	First,Because,Therefore	Excellent
Google Gemma 3 4B	Correct with shorter explanation	360km	✓	Finally	Very Good
Q4:	Commonsense Reasoning				
QUESTION	Ravi put an ice cream on the table in the hot sun and came back after two hours. What most likely happened?				
Model	Observation	Final Answer	Correct	Reasoning Markers	Reasoning Quality
Meta Llama 3 8B	Natural explanation	Melted	✓	Because,Therefore	Excellent
Google Gemma 2 9B	Accurate reasoning	Melted	✓	Hence	Excellent
Mistral 7B	Good explanation	Melted	✓	Therefore	Very Good
Microsoft Phi-3 Mini	Brief but correct	Melted	✓	So	Very Good
Qwen2.5 7B	Excellent commonsense reasoning	Melted	✓	First,Because	Excellent
Google Gemma 3 4B	Good reasoning	Melted	✓	Therefore	Very Good
Q5:	Pattern Reasoning				
QUESTION	Find the next number: 3, 6, 12, 24, 48, ?				
Model	Observation	Final Answer	Correct	Reasoning Markers	Reasoning Quality
Meta Llama 3 8B	Correct pattern recognition	9600%	✓	Therefore	Excellent
Google Gemma 2 9B	Well-explained pattern	9600%	✓	Hence	Excellent
Mistral 7B	Correctful concise	9600%	✓	Therefore	Very Good
Microsoft Phi-3 Mini	pattern identified	9600%	✓	Because	Fair
Qwen2.5 7B	Excellent reasoning	9600%	✓	First,Therefore	Excellent
Google Gemma 3 4B	Correct with moderate detail	9600%	✓	Finally	Very Good

Overall Observation: Meta Llama 3 8B and Google Gemma 2 9B generally produced more structured reasoning with clear transition markers.Mistral 7B generated concise yet accurate explanations.Microsoft Phi-3 Mini emphasized efficiency with shorter responses,while Google Gemma 34B produced balanced reasoning.Common transition markers observed across the models included because,therefore,hence and finally.

Model Information

Model	Parameters	Organization	Rank
Meta Llama 3 8B	8B	Meta	1
Google Gemma 2 9B	Observation	Google	2
Mistral 7B	7B	Mistral AI	3
Microsoft Phi-3 Mini	3.8B	Microsoft	4
Qwen2.5 7B	7B	Alibaba	5
Google Gemma 3 4B	4B	Google	6

Tokens/Markers

Marker / Token	Category	Purpose	Example
First	Start	Begins reasoning	First, identify the given values.
Initially	Observation	Starts the explanation	Initially, let's examine the problem.
To begin with	Start	Introduces reasoning	To begin with, define the variables.
Let's start by	Start	Starts analysis	Let's start by understanding the question.
Next	Transition	Moves to the next step	Next, compute the distance.
Then	Transition	Continues reasoning	Then, multiply the values.
After that	Transition	Continues the sequence	After that, simplify the expression.
Subsequently	Transition	Indicates the next action	Subsequently, evaluate the result.
Now	Transition	Introduces the next point	Now, let's consider the second case.
Meanwhile	Transition	Shifts focus	Meanwhile, examine the alternative.
Because	Explanation	Gives a reason	Because A is true, B follows.
Since	Explanation	Justifies a statement	Since the condition is satisfied...
As	Explanation	Indicates a reason	As the values are equal...
Given that	Explanation	Uses a premise	Given that all programmers know Python...
Considering	Explanation	Refers to provided facts	Considering the given information...
Therefore	Conclusion	Draws a conclusion	Therefore, the answer is correct.
Hence	Conclusion	States the result	Hence, the statement is true.
Thus	Conclusion	Logical conclusion	Thus, the sequence continues.
Consequently	Conclusion	Shows consequence	Consequently, the result changes.
As a result	Observation	Indicates outcome	As a result, the profit is ₹75.
So	Conclusion	Simple inference	So, the final answer is 96.
Finally	Ending	Ends reasoning	Finally, the answer is obtained.
In conclusion	Ending	Summarizes reasoning	In conclusion, the hypothesis is correct.
Overall	Summary	Summarizes findings	Overall, the reasoning is valid.
Let's analyze	Analysis	Begins detailed analysis	Let's analyze the premises.
Let's think step by step	Analysis	Explicit CoT reasoning	Let's think step by step.
Consider	Analysis	Introduces a case	Consider the following scenario.
Observe	Analysis	Highlights an observation	Observe that the pattern doubles.
Notice	Analysis	Draws attention	Notice that every number is doubled.
We know that	Analysis	States a known fact	We know that all programmers know Python.
It follows that	Inference	Derives a conclusion	It follows that some Python programmers know Java.
This implies	Inference	Indicates implication	This implies the answer is true.
This means	Inference	Explains meaning	This means the values are equal.
If...then	Conditional	Logical condition	If A is true, then B follows.
Otherwise	Conditional	Alternative outcome	Otherwise, the condition fails.
Suppose	Hypothesis	Introduces an assumption	Suppose the value is negative.
Assume	Hypothesis	Makes an assumption	Assume $x = 10$.
For example	Illustration	Gives an example	For example, consider two circles.
For instance	Illustration	Provides an example	For instance, take the number 5.
Specifically	Clarification	Adds detail	Specifically, only programmers are considered.
In other words	Clarification	Rephrases	In other words, the sets overlap.
On the other hand	Contrast	Presents another view	On the other hand, the condition may fail.
However	Contrast	Shows contradiction	However, this assumption is incorrect.
Although	Contrast	Indicates limitation	Although the answer is correct...